

Heart Disease Analysis

Phase 5: Project Development Phase

Preprocessing Steps & Business Questions with Visualization

Naga Nandini Padidham | [GitHub: nandinipadidham-lgtm/Heart-Disease-Analysis](#)

Part A: Data Preprocessing Steps

This section documents how the raw dataset was taken from its source to an analysis-ready state in MySQL.

1. Data Source

- Dataset: Heart Disease Dataset (Kaggle)
- Format: CSV (Heart_new2.csv)
- Size: 4,500 records, 18 columns

2. Data Cleaning

- Checked for missing/null values across all 18 columns.
- Verified consistent categorical labels (e.g. 'Yes'/'No' fields, Sex as Male/Female).
- Confirmed numeric fields (BMI, PhysicalHealth, MentalHealth, SleepTime) were stored as valid numeric types.
- Standardized the Diabetic column, which includes 4 distinct values: Yes, No, No borderline diabetes, and Yes (during pregnancy).

3. Loading into MySQL

```
CREATE DATABASE heart_disease_db; USE heart_disease_db; SELECT COUNT(*) FROM HD;
```

The cleaned CSV was imported into a table named HD inside the heart_disease_db database. Row count was verified at 4,500 to confirm a complete, lossless import.

4. Derived / Bucketed Fields Created in SQL

To make analysis and visualization more meaningful, several continuous fields were bucketed using SQL CASE statements:

```
-- Sleep duration buckets CASE          WHEN SleepTime < 6 THEN 'Under 6h'          WHEN  
SleepTime BETWEEN 6 AND 8 THEN '6-8h'          ELSE 'Over 8h' END AS sleep_bucket -- BMI  
category buckets CASE          WHEN BMI < 18.5 THEN 'Underweight'          WHEN BMI < 25 THEN  
'Normal'          WHEN BMI < 30 THEN 'Overweight'          ELSE 'Obese' END AS bmi_category
```

These buckets made it possible to compare heart disease rates across meaningful groups rather than raw continuous values.

5. Preparing Data for Tableau

- Connected Tableau Desktop to the MySQL heart_disease_db database.
- Converted the live MySQL connection into a Tableau Extract, since Tableau Public does not support live database connections.
- Verified field types (dimensions vs measures) were correctly interpreted by Tableau.

Part B: Business Questions with Visualization

Each business question below was answered using a specific SQL query and/or Tableau visualization.

Business Question	Visualization Used	Key Finding
What percentage of patients have heart disease?	KPI card + pie chart (Heart Disease Distribution)	10.7% (483 of 4,500)
Does heart disease risk increase with age?	Bar chart — Heart Disease by Age Category	Rises steadily; peaks at 60-69
Is heart disease more common in men or women?	Bar chart — Heart Disease by Gender	Slightly higher in males
Does race show different heart disease rates?	Bar chart — Heart Disease by Race	Rate varies; White and Black populations largest sample sizes
Are smokers more likely to have heart disease?	Grouped bar — Smoking vs Alcohol Drinking	Smokers show higher rate than non-smokers
Is diabetes linked to heart disease?	Bar chart — Diabetes vs Heart Disease	Diabetics show markedly higher rate
Is stroke history linked to heart disease?	Bar chart — Stroke vs Heart Disease	Stroke patients show much higher rate
Does sleep duration relate to heart disease?	Scatter plot — Sleep Time Distribution by Age	Heart disease cases rise in older age groups regardless of sleep
Does self-rated general health predict heart disease?	Bubble chart — General Health Analysis	'Poor'/'Fair' health correlates with higher rate
Which combined patient profile is highest risk?	SQL: multi-column GROUP BY (Age + GenHealth + Diabetic)	Older, poor-health, diabetic patients rank highest

Summary

Together, these ten business questions map directly onto the dashboard's KPI cards and charts, and onto the six-part Tableau Story — ensuring every visualization in the project answers a specific, stated question rather than existing as decoration.